Chapter 5 Quiz — Collisions Transfer Energy

Grade 4 · Energy Transfer and Transformation

Name: _____ Date: _____

Circle or write the best answer.

PART 1 — Multiple Choice

1. A collision happens when two objects ___.

a) disappear b) come together with impact c) change color d) float

2. When a moving car hits a still car, what happens to some of the moving car's energy of motion?

a) it disappears b) it transfers to the still car c) it turns into nothing d) it stops existing

3. The SAME car rolls faster into a target. What happens?

a) the target moves LESS b) the target moves MORE c) nothing changes d) the car disappears

4. Two cars roll at about the SAME speed into the same target. One car weighs more. Which car moves the target farther?

a) the heavier car b) the lighter car c) both the same d) neither

5. In a collision, some energy of motion becomes ___ energy that we hear.

a) light b) sound c) stored d) electric

6. What NEVER happens to energy in a collision?

a) it transfers b) it changes form c) it disappears d) it moves to the target

PART 2 — The Model

7. Fill in the missing step of the collision model:

COLLISION → MOTION CAN CHANGE → ENERGY CAN TRANSFER → _____

PART 3 — Data

8. A student rolls the SAME car down three ramp heights and measures target displacement.

Ramp height	Average target displacement (cm)
1 book	13
2 books	26
3 books	40

At which ramp height did the car move the target the farthest? What does this show about speed and energy?

PART 4 — Fair Test

9. Which is a fair test of how SPEED affects a collision?

a) change the car AND the ramp b) change only the ramp height (same car) c) change only the car (same ramp) d) change everything

PART 5 — Short Explanation

10. A car hits a cup. The cup moves and we hear a sound. Write where the energy went. Use the model: “I know the energy did not disappear because ___.”